PRODUCT PERFORMACE UPDATE

Application: W Abrasives, Ontario Canada – Retort Furnace

RETORT FURNACE DETAILS

- **Furnace:** Temper shot furnace #1 is 17 feet long inside refractory and approximately 48 inches in diameter. There are nine (9) burners total each rated at 450,000 Btu.

- **Lining Configuration:** The current ling consists of two layers of IFB, the hot face is 9" of 2300^0 IFB and the backup lining is 3" of 2000^0 IFB.

- **Customer Need:** ITC recommended a solution to address several major challenges,
  - High fuel consumption.
  - Continual refractory maintenance and repair.

ITC RECOMMENDATION

- **Engineered Approach:** ITC recommended a hot face coating to address the customers' requirements. The coating design included a proprietary hot face coating, ITC 100HT, applied to the hot face of the IFB.

- **High Temperature, Energy Efficient Ceramic Coatings:** ITC manufactures several proprietary, high temperature, energy efficient ceramic coatings to work in concert with refractory or metal substrates. Under ideal conditions, high these coatings absorb energy from the process and re-radiate it to the load, imparting the following benefits;

  Lower heat loss – Minimizes heat transfer through the ladle preheat lid refractory
  Lower Maintenance costs – Protects the substrate refractory
  Reduced preheating time – Improves the thermal efficiency of the process thus allowing quicker furnace start-up after downturns.

IMPACT ON THE PROCESS

- REFRACTORY LONGEVITY
  Current refractory maintenance costs are estimated at $10,000.00 per year. With the use of ITC Coatings refractory life is extended on average 2 to 3 times longer

- REDUCED PREHEAT TIME

- IMPROVED FUEL SAVINGS
  Due to the thermal efficiencies, kWh savings was reduced by 7% - 10%

- REDUCED ENERGY CONSUMPTION
  Before ITC coating, average fuel consumption was 325kWh/ton
  After ITC coating, average fuel consumption is now 300 kWh/ton

- REDUCED AMOUNT OF CO_2 & NO_X EMITTED
  By increasing the thermal efficiency of the furnace and lowering the gas consumption, more oxygen is burned thus lowering the amount of pollutants emitted.

IMPACT ON BOTTOM LINE COSTS

- **FUEL SAVINGS OF 7% - 10%**
  Before:
  16,000 ton/yr x 325 kWh/ton x $0.20/m^3 x m^3/10.44 kWh = $99,617.00/yr

  After:
  16,000 ton/yr x 300 kWh/ton x $0.20/m^3 x m^3/10.44 kWh = $91,954.00/yr

- **CO_2 REDUCTION OF 8%**
  Before:
  16,000ton/yr x 325 kWh/ton x therm/29.3 kWh x 0.005302 ton CO_2/therm = 940 ton CO_2/yr

  After:
  16,000ton/yr x 300 kWh/ton x therm/29.3 kWh x 0.005302 ton CO_2/therm = 869 ton CO_2/yr

- **ROI** = 2.4 months

TOTAL IMPACT/YEAR = $ 17,633.00